Working in iframes
WebDriver - Lecture - 9

Agenda

- Handling **iframes**
- Notes Script in Box Portal
- Dynamic Waiting for New Tab
- Detecting **iframe** Elements
- Dynamic **Switching** to **iframes**

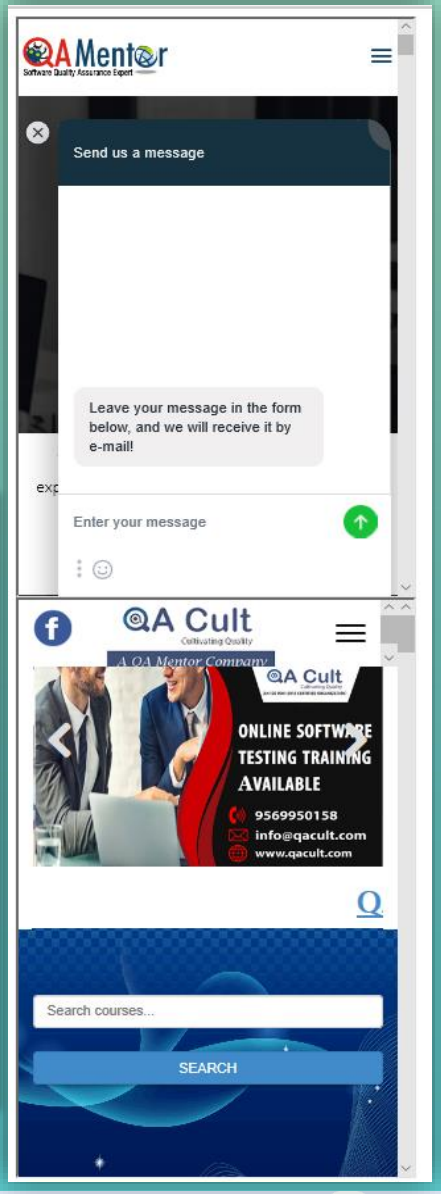
- Renaming the Note
- **Switching Back** from Frames
- Deleting the Note
- Quiz
- Homework

Handling iframes

- **iframe** is a **web page** which is **embedded** in **another web page** or an **HTML document** embedded inside **another HTML document**.
- An **iframe tag** is defined using **<iframe></iframe>** tags.
- **TargetLocator interface** helps in **switching** from **main window** to **iframe** and **back**
- **Switching** to **iframes** can be done in <u>3 ways</u>:
  - **By Index -** Index of the **iframe** starts with **'0'**. Suppose if there are **10 frames** in a page, we can **switch** to the **iframe** by using index.
    - *driver.switchTo().frame(0);*
    - *driver.switchTo().frame(1);*
  - **By Name or Id –** We can switch to **iframe** using name or id attribute of **iframe WebElement**.
    - *driver.switchTo().frame("name of the element");*
    - *driver.switchTo().frame("id of the element");*
  - **By Web Element –** We can switch to **iframe** using **web element**.
    - *driver.switchTo().frame(WebElement);*

<u>**Note**</u>**: frame()** method of **TargetLocator** returns **WebDriver** object.

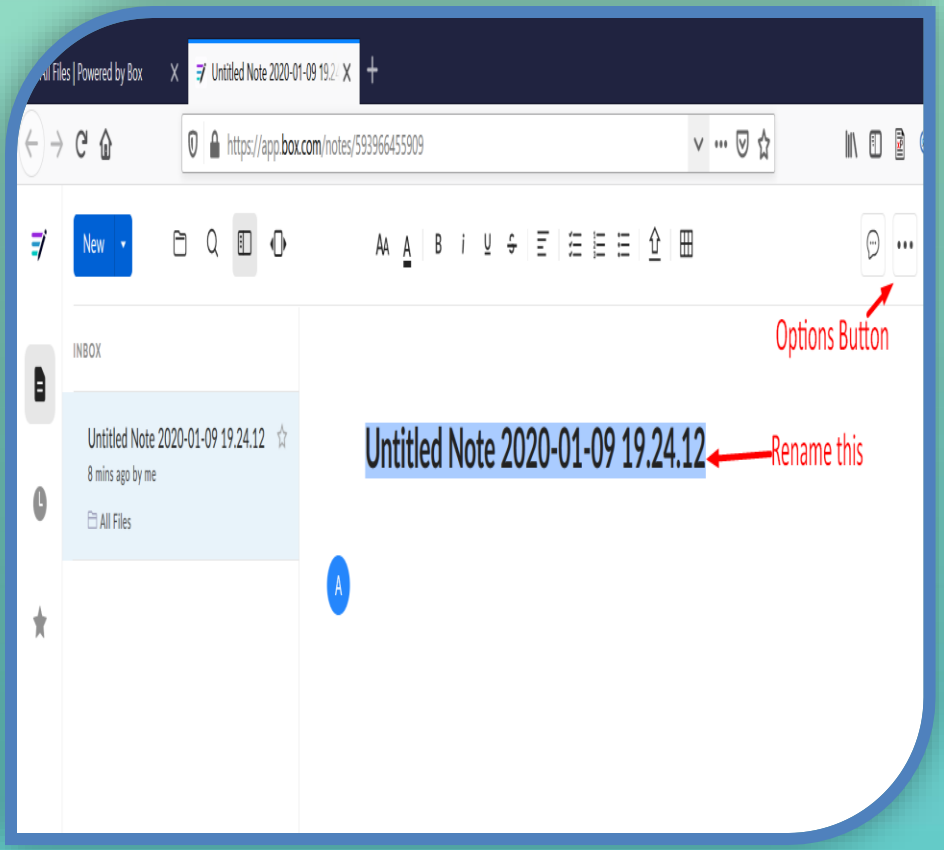
```html
<html>
  <body>
    <div class="box">
      <iframe name="iframe1" id="IF1"
      height="600" width="400"
      src="http://qamentor.com"></iframe>
    </div>
    <div class="box">
      <iframe name="iframe2" id="IF2"
      height="600" width="400"  align="left"
      src="http://qacult.com"></iframe>
    </div>
  </body>
</html>
```



Notes Script in Box Portal

WebDriver script for **creating and deleting** note in Box portal:

- Navigate to http://app.box.com
- **Login** into app
- Verify the title of **"All Files"** page
- Get the **handle** of current window **("All Files")** page
- Click on **"Notes"** button in left panel of **"All Files"** page
- **Verify new tab** is opened
- **Switch Window** and **verify title** of new tab is **"Box Notes"**
- **Switch** to **first iframe** and click on **new** dropdown to **create new note**
- **Switch** to **second iframe** within **first iframe**
- **Rename** the newly created **note**
- Switch back to **first iframe**
- Click on **Options button** and **Delete the note**
- **Close** the "**Box Notes**" **window**
- **Logout** from the app in **main window**



Dynamic Waiting for New Tab

- Whenever you open a new tab, number of **window handles** increases its **size** by **1**
- Here, for **Notes script** we have to click on **"Notes" button** in the left panel of **"All Files"** page which will open new **WebDriver** tabbed window
- For working with this **new tabbed window**, we have to **switch** to it
- But clicking on **Notes button** takes some time to increase the size of window handles by 1. We should **wait dynamically** for the window handle size to increase with ***numberOfWindowsToBe()*** *method*

```
int windowSize = driver.getWindowHandles().size();
notesBtn.click();
// ***********************************************
wait.until(ExpectedConditions.numberOfWindowsToBe(windowSize + 1));
// ***********************************************
```

- In the above snippet:
  - Take the **size of window handles** in a variable before clicking button to open new tab
  - Then after click, wait for **window size to increase by 1** with ***numberOfWindowsToBe()*** *method*
  - **ExpectedCondition<Boolean> numberOfWindowsToBe(int expectedCount) –** Returns true if expected number of window count is achieved within timeout duration

Detecting iframe Elements

- **Detecting** of **iframe elements** has become very easy with **Chrome new Developer tools**.

- **Earlier** it was **not possible** with Chrome and only **Firefox browser** is used.

- It is still recommended to use Firefox browser for **detecting iframe elements**.
  - Inspect **iframe element** in main window.
  - Copy the **url** from the **"src"** attribute of given **iframe**.
  - **Open new tab** and **paste** the **url**.
  - This opens the content of **HTML within iframe** in new **tabbed window**. (This technique works only when "**src**" attribute is given in **iframe** element)
  - If **"src"** attribute in **iframe** is not provided, then we **can not** open **HTML of iframe** as separate window.

```
<!DOCTYPE html>                                   Main HTML
<html lang="en-US"> event
  ▶ <head> ··· </head>
  ▼ <body>
    ▼ <iframe id="service_iframe" name="service_iframe" allowfullscreen="true"      copy "src" attribute
      src="https://notes.services.box.com/p/note?authCode=4MRlRSzYlvD4qAk99gBTICUF8NwRtZti&hostname=app.box.com"
      width="100%" height="100%" frameborder="0">
      ▼ #document
          <!DOCTYPE html> iframe HTML
        ▼ <html> event
            ▶ <head> ··· </head>
            ▼ <body id="mod-page-notes-1" class="box-notes left-sidebar-opened disable-page-element-transition"
              data-module="page-notes" data-view-state="edit" data-banner-type="connected">
              ▶ <div id="image-overlay" class="hidden" tabindex="0" data-module="image-overlay">···</div> event
                <div id="feature-splash-container" class="react-ui" data-module="feature-splash"></div>
```

Dynamic Switching to iframes

- Before working with **frame elements**, you have to **switch** to the **frame** with the **name** or **id** of the **frame** element or using its **index** or **WebElement**.
- Its always a good practice to identify total number of iframes on a webpage by using below snippet.

```
int frameSize = driver.findElements(By.tagName("iframe")).size(); // total iframes on page
```

- **TargetLocator** interface is used to switch to iframe using:
  - **driver.switchTo().frame("iframeName");**
- But now **switch** to **iframe** after **waiting** for it **dynamically** using **explicit wait's condition** ***frameToBeAvailableAndSwitchToIt***() method.
- Code Snippet:

```
// ****************************************
wait.until(ExpectedConditions.frameToBeAvailableAndSwitchToIt("service_iframe"));
// ****************************************
```

- **ExpectedCondition<WebDriver> frameToBeAvailableAndSwitchToIt(String nameOrId)** – This method waits for the **frame** to be **available** using its **name**, **id**, **index**, **By locator** or **WebElement**.
  - If **frame** is available to **switch** within **timeout duration**, it **switches** to the **frame** and returns **WebDriver** instance.
  - Otherwise, if **frame** is <u>not found</u> within **timeout duration**, it throws **Timeout exception**.

Renaming the Note

- **Renaming** of newly created **note** from **Untitled Note** to some **new value** involves **switching** to another **iframe**.
- Without **switching** to **inner iframe**, you will not be able to rename the note.
- When **renaming** the note to some **new value**, you have to look for element with **"input"** tag because that can accept **String value** with **sendKeys()** method.
- Code Snippet For **Renaming**:

```
int frameSize1 = driver.findElements(By.tagName("iframe")).size();
System.out.println("Total second Frames --" + frameSize1);
wait.until(ExpectedConditions.frameToBeAvailableAndSwitchToIt(1));
```

Switching to inner frame with index

Renaming the note

```
wait.until(ExpectedConditions.presenceOfElementLocated(By.id("documentTitle"))).click();
wait.until(ExpectedConditions.presenceOfElementLocated(By.id("titleEditor"))).sendKeys("notestest4");
```

- **Renaming the note** involves:
  - Click on the **header title** of the **note** (something like **Untitled Note**) is required **before sending keys** to **textfield** which is otherwise **dead** and **clicking** on header **enables** this **textfield**.
  - After **click** on header **element**, send some **new value** to **textfield** box, thus **renaming** the note.

Switching Back from Frames

- **TargetLocator** interface also provides methods to switch back.
- There are **two methods** for switching back:
  - **WebDriver parentFrame()** – With this method you can **switch back** to outer **frame** from **inner frame** or **returns** the **driver** of parent frame.
  - **WebDriver defaultContent()** – With this method you can **switch back** to **main window** which has **iframes** or **returns** the **driver** of top window.

```
driver.switchTo().parentFrame();      //switch back to parent frame
driver.switchTo().defaultContent();//switch back to top window
```

Deleting the Note

- **Deleting** the note created involves **switching back** to parent frame with **parentFrame()** method.
- Delete the note
  - Click on **options button**
  - Click on "**Delete this note**" to delete
- After deleting **close** the **"Box Notes" window**.

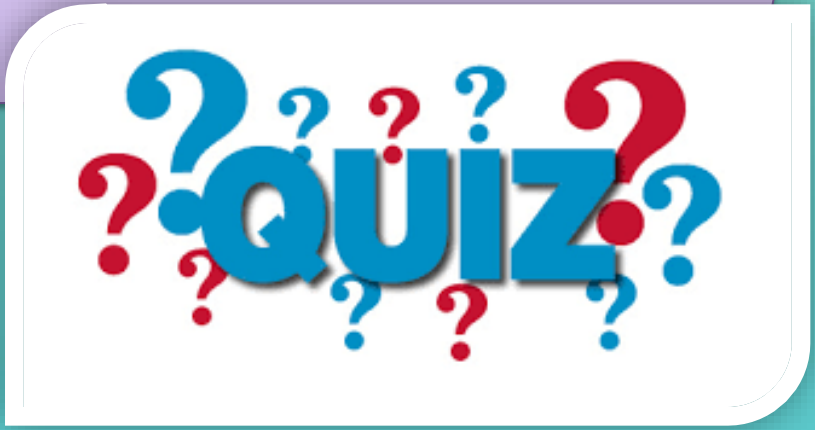
```
driver.switchTo().parentFrame();                                    Deletion Operation
wait.until(ExpectedConditions.presenceOfElementLocated(By.xpath("//div[@id='options-menu-editbar']"))).click();

wait.until(ExpectedConditions.visibilityOfElementLocated(
        By.xpath("//div[@id='options-menu-editbar']//a[contains(@class,'options-menu-delete')]"))).click();
driver.close();        //closes the "Box Notes" tab
```

- **close()** and **quit()** method of **WebDriver**:
  - **void close() – Closes** the current window which has focus.
  - **void quit() - Quits** the **WebDriver**, closing every **associated window** and **ends the session**.
- **getPageSource()** method of **WebDriver**:
  - **String getPageSource()** – Gets the **page source** as String of **the current focused window**, means the complete **HTML DOM.**

Quiz

- What is **iframe** in **HTML** and which interface is required to switch to **iframe**?
- What are the **3 ways** to switch to **iframe**?
- How will you **dynamically** wait for **window handles** to increase its size?
- How will you **dynamically** wait and switch to **iframe**?
- Name **two methods which** are used to **switch back** from **iframe** and explain their **usage**?
- What is the difference between **close()** and **quit()** methods of **WebDriver?**



Homework

- Complete **WebDriver** script to create and delete "**Box Note**".
  - After deleting, **close** the "**Box Note**" Tab
  - In the first tab on "**All Files**" page, **logout** from the portal.
  - **Quit** the webdriver

- Create another WebDriver script:
  - Navigate to http://qacult.com
  - Click on "**All Events**"
  - From there, start **first video** (you have to **switch** to **iframe**)